

Nicole Thiessen

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering

Waterloo, ON

2025 - 2030

TECHNICAL SKILLS

Electrical: Battery Systems, High-Voltage Safety, Soldering, Wire harnessing, HIL Bench testing, CAN bus analysis

Mechanical: SolidWorks, AutoCAD, GD&T, CFD Simulations, DFM

Software: C++, Python, JavaScript, CANalyzer, Git, TensorFlow, PyTorch, OpenCV

EXPERIENCE

Vehicle Systems Engineering Intern

May 2026 – August 2026

Geotab

Waterloo, ON

- Reverse-engineered proprietary CAN signals in vehicles using CANalyzer, Vector hardware, and scan tools, expanding Geotab's telematics support for EV fleets
- Validated CAN signals through vehicle testing and remote fleet analytics, improving telematics data quality for customer applications
- Performed HIL bench simulations to emulate vehicle-device interaction, testing telematics firmware before deployment across global vehicle fleets

Technical Analysis and Intellectual Property Assessment Intern

June 2025 – August 2025

CoreCurrent Solutions Inc.

Ottawa, ON

- Reverse engineered automotive electronic modules such as ADAS Controls and Front Radar Sensors to identify patent infringements, accelerating legal evidence collection
- De-soldered and researched IC components, compiling findings into organized reports, improving accuracy in client cases
- Built a Python OpenCV tool mapping chip images to their positions on the PCB, streamlining analysis workflows

PROJECTS

Formula Electric Design Team - Battery Electronics Responsible Engineer

Sept 2025 - Present

- Integrated the battery packs' high-voltage wire harness and low-voltage safety systems, while leading the redesign of next year's battery electronics architecture
- Developed a cell-grouping algorithm for battery segments, using cell DCIR data to balance power output and long-term battery health
- Manufactured lithium-ion battery segments through epoxy bonding, resistance spot welding, and AMS board assembly

AI Voice Assistant

August 2025

- Built an AI voice assistant using Whisper and LLM Dialogue, enabling real-time conversations
- Modeled and rigged a 3D robot avatar that reacted to emotions. Designed a custom neural network using PyTorch to map emotion metrics to joint angles, delivering human-like AI interaction

Synthetic Data Generator

July 2025

- Built a synthetic data generator to solve AI training data scarcity, with a pipeline that combines CTGAN generation and LLM enrichment to create large (10k+ samples) and flexible datasets

UW Crowd – Hack the North

Sept 2025

- Developed real-time campus occupancy website using Raspberry Pis' to scan for Bluetooth Low Energy signals, helping students plan around crowds.

AWARDS & ACHIEVEMENTS

Academic: 4.0 GPA, 43/45 score on the IB Diploma

Governor General Academic Bronze Medal: Highest academic average, 250+ students

Colonel By S.S. Graduate Award: Academic, athletic, and leadership excellence